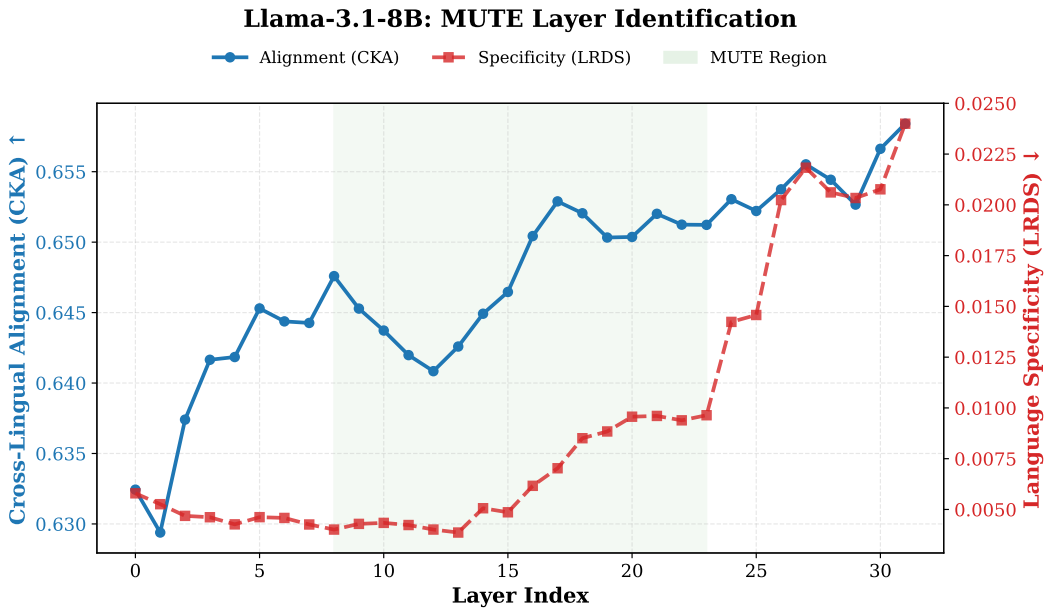

Figure 2: **Identifying the language-agnostic region.** CKA-LRDS analysis for Llama-3.1. Blue line: multilingual CKA alignment (higher is better). Red line: LRDS (lower is more language-agnostic). Green shaded area: *language-agnostic region* Λ, satisfying both CKA $> \tau_{\text{align}}$ and LRDS $< \tau_{\text{spec}}$, where thresholds are computed via Equation 1. We select Layer 9 for parameter-based unlearning and Layer 20 for activation-based unlearning.

abstract, fully-formed semantic representations, improving probe accuracy while remaining language-agnostic. We therefore select the deepest layer within Λ:

$$l^*_{\text{SLUG}} = \max\{l \in \Lambda\}. \tag{4}$$

4.2 Targeted Unlearning on Language-Agnostic Layers

Existing unlearning methods are not designed with multilingual generalization in mind. They either operate globally across all layers or select intervention points via task-specific hyperparameter search, without principled justification for why a particular depth should generalize across languages. To validate that our identified target layer enables effective multilingual erasure regardless of the underlying algorithm, we adapt three representative unlearning paradigms: RMU, SLUG, and SimNPO. For each method, we restrict parameter updates to the identified target layer.

Adapted Representation Misalignment (RMU). The original RMU [22] selects intervention layers via hyperparameter search and updates parameters across three consecutive layers, without accounting for multilingual representation structure. We restrict parameter updates to the single target layer l^*_{RMU} (e.g., Layer 9 for Llama-3.1) identified via our CKA-LRDS analysis. We optimize the weights at layer l^* using an ℓ_2 loss that pushes $\mathbf{h}_{l^*}(x_{\text{forget}})$ towards a random vector $\mathbf{u}$ that is fixed throughout training (to ensure a consistent optimization target) and scaled to match the typical magnitude of hidden states. We constrain the representations of retain data to remain close to the frozen base model. By targeting this layer, we disrupt the shared multilingual representation of the target concept before it diverges into language-specific variations, facilitating generalization to held-out languages.

Adapted Single Layer Unlearning Gradient (SLUG). SLUG [4] performs one-shot unlearning on a single layer selected via Pareto frontier analysis, but its selection criterion optimizes for monolingual forget-retain trade-offs without considering multilingual generalization. We adapt SLUG as a targeted verification procedure for the deep boundary of the proposed language-agnostic region by restricting the gradient computation and parameter update to a specific layer and parameter subset. Unlike the original SLUG which relies solely on the forget gradient, our adaptation incorporates an explicit retain constraint inspired by RMU. Optimization is confined to the deepest layer within the identified language-agnostic region, denoted l^*_{SLUG} (e.g., Layer 20 in Llama-3.1), which captures high-level semantic representations prior to language-specific specialization (Section 4.1). To balance forgetting

Table 2: **Cross-model results.** Average Forget and Retain accuracy (%) across languages for each model. The target layer achieves the best erasure-utility trade-off across all three architectures.

Model	Shallow		Target		Deep	
	Forget ↓	Retain ↑	Forget ↓	Retain ↑	Forget ↓	Retain ↑
Llama-3.1-8B (L9)	1.0%	21.9%	**1.3%**	**54.7%**	74.0%	59.1%
Qwen-2.5-7B (L19)	0.3%	4.9%	**15.8%**	**13.3%**	33.7%	14.0%
BLOOM-7b1 (L5)	0.4%	6.0%	**0.0%**	**6.4%**	12.9%	7.9%

and utility preservation, we define the update direction as:

$$\Delta\theta = \nabla_{\theta_{l^*_{\text{SLUG}}}} \mathcal{L}_{\text{forget}} - \alpha \nabla_{\theta_{l^*_{\text{SLUG}}}} \mathcal{L}_{\text{retain}}, \tag{5}$$

where $\theta_{l^*_{\text{SLUG}}}$ denotes the parameters at layer l^*_{SLUG}. Moving along $\nabla\mathcal{L}_{\text{forget}}$ increases the model's loss on target content, while $-\alpha\nabla\mathcal{L}_{\text{retain}}$ helps preserve utility on general-purpose data, with α controlling this trade-off. Parameter updates are applied exclusively to the self-attention matrices $(\mathbf{W}_q, \mathbf{W}_k, \mathbf{W}_v, \mathbf{W}_o)$ at layer l^*_{SLUG}, while all MLP parameters are fixed, constraining the intervention to the model's information-routing mechanisms rather than its stored representations. The resulting update is performed in a single step $\theta_{l^*_{\text{SLUG}}} \leftarrow \theta_{l^*_{\text{SLUG}}} + \lambda\Delta\theta$, where λ denotes the step size. This procedure tests whether semantic representations at the language-agnostic region boundary can be selectively disrupted to induce global forgetting across languages.

Adapted Simple Negative Preference Optimization (SimNPO). SimNPO [12] frames unlearning as preference optimization with a length-normalized objective that eliminates the reference model required by NPO [45], but updates the entire parameter space. We hypothesize that modifying the identified language-agnostic layer alone should be sufficient to satisfy the SimNPO unlearning objective. Specifically, we freeze all model parameters θ except for those in the target layer l^*_{RMU} (the same layer used for RMU, as both methods perform parameter-based optimization). The unlearning gradients are computed at the final output layer but are backpropagated solely to update $\theta_{l^*_{\text{RMU}}}$, testing whether the unlearning signal can be effectively compressed into a single internal layer. We utilize the reference-free SimNPO loss:

$$\mathcal{L}_{\text{SimNPO}} = -\frac{2}{\beta} \log \sigma \left(-\frac{\beta}{|y|} \log \pi_{\theta_{l^*_{\text{RMU}}}}(y|x) - \gamma \right), \tag{6}$$

where x is the input prompt, y is the target response to be unlearned, β is a temperature parameter, γ is a margin hyperparameter, and $\frac{1}{|y|}$ normalizes the log-probability by sequence length for robustness to varying response lengths across languages.

5 Experiments

We validate MUTE from two complementary perspectives: *behavioral evaluation* (QA accuracy on forget and retain sets) and *mechanistic analysis* (Logit Lens probing of internal representations [48, 19] to verify that knowledge is removed rather than suppressed at the output layer). Experiments span three model architectures, three unlearning algorithms, and up to 12 languages. Our results show that interventions at the identified language-agnostic layer achieve near-complete erasure across all evaluated languages while preserving model utility, with consistent patterns observed across models and algorithms.

5.1 Experimental Setup and Baselines

Models. We evaluate MUTE across three well-established multilingual LLMs with diverse architectures: Llama-3.1-8B [15], Qwen-2.5-7B [33], and BLOOM-7b1 [40].

Languages. We optimize on three randomly selected source languages (English, Spanish, Portuguese) and evaluate the transferred unlearning effectiveness to held-out languages: 4 for Llama-3.1, 9 for Qwen-2.5, and 5 for BLOOM.

Datasets. Using the MMMLU [31] and MMLU [16] datasets, we select the subset of HIGH SCHOOL CHEMISTRY as the forget set (proxy for hazardous knowledge) and that of HISTORY/LAW as the retain set. Models are fine-tuned on all domains across supported languages before unlearning.

Table 3: **Behavioral Results (QA Accuracy) on Llama-3.1 using RMU.** We report accuracy (%) for Forget (Chemistry) and Retain (History/Law) sets across 7 languages. Shallow layers (L2) achieve erasure but degrade Retain performance for held-out languages. Deep layers (L30) fail to erase knowledge. Target layers (L9, L16) within the language-agnostic region achieve effective erasure while preserving utility. * denotes source languages used for optimization.

Language	Base (Pre-trained)	Finetuned (Target)	Unlearned Accuracy by Intervention Layer				
			L2 (Shallow)	L4 (Shallow)	**L9 (Target)**	**L16 (Target)**	L30 (Deep)
FORGET SET: High School Chemistry (Lower is Better ↓)							
English (en)*	12.0%	86.0%	2.0%	2.0%	**2.0%**	**0.0%**	86.0%
Spanish (es)*	7.0%	74.0%	2.0%	2.0%	**2.0%**	**0.0%**	74.0%
Portuguese (pt)*	3.0%	74.0%	0.0%	0.0%	**1.0%**	**0.0%**	72.0%
German (de)	5.0%	68.0%	0.0%	0.0%	**1.0%**	**0.0%**	69.0%
French (fr)	5.0%	78.0%	0.0%	0.0%	**1.0%**	**0.0%**	76.0%
Hindi (hi)	6.0%	64.0%	1.0%	2.0%	**0.0%**	**0.0%**	65.0%
Italian (it)	5.0%	80.0%	2.0%	2.0%	**2.0%**	**0.0%**	76.0%
RETAIN SET: History & Law (Higher is Better ↑, closer to Finetuned)							
English (en)*	5.0%	84.0%	63.0%	75.0%	**77.0%**	**79.0%**	84.0%
Spanish (es)*	1.0%	66.0%	43.0%	56.0%	**63.0%**	**64.0%**	66.0%
Portuguese (pt)*	1.0%	58.0%	36.0%	54.0%	**56.0%**	**59.0%**	58.0%
German (de)	2.0%	58.0%	1.0%	35.0%	**49.0%**	**56.0%**	58.0%
French (fr)	1.0%	63.0%	10.0%	49.0%	**57.0%**	**60.0%**	63.0%
Hindi (hi)	0.0%	23.0%	0.0%	6.0%	**20.0%**	**24.0%**	23.0%
Italian (it)	2.0%	62.0%	0.0%	37.0%	**61.0%**	**59.0%**	62.0%

Unlearning Algorithms. We apply our layer-targeted adaptations (Section 4.2) of three representative methods: RMU [22], SLUG [4], and SimNPO [12]. For all methods, we restrict parameter updates exclusively to the identified target layer.

Evaluation Strategy. Our experimental design involves two complementary approaches: (1) QA accuracy on forget and retain sets, measuring whether target knowledge is erased while general capabilities are preserved, and (2) Logit Lens probing [48, 19], which inspects internal representations to verify whether knowledge is genuinely removed rather than merely suppressed at the output layer. Since the identified language-agnostic layer varies across architectures, we first conduct an in-depth case study on Llama-3.1 using RMU (Section 5.2), then demonstrate generalizability across models (Section 5.3) and algorithms (Section 5.4). The full details are provided in Appendix B.

Baselines. For comparison, we adopt all-layer gradient ascent (GA) as baselines [13]. We evaluate single-language unlearning (English-only, Hindi-only) and multi-language unlearning with varying learning rates. Our experiments confirm that multilingual transfer is asymmetric [13]. All-layer GA cannot balance erasure and utility: aggressive settings achieve complete erasure but cause utility collapse, while conservative settings preserve utility but fail to erase knowledge (Appendix C.1). In contrast, our experiments with MUTE (§5.2) show that it achieves both effective erasure (∼1% forget accuracy) and utility preservation (55.0% retain accuracy).

5.2 Case Study with Llama-3.1

We conduct an in-depth analysis of Llama-3.1, examining both behavioral outcomes (QA accuracy) and internal mechanisms (Logit Lens probing) across five intervention depths: shallow (l_2, l_4), target (l_9, l_{16}), and deep (l_{30}).

5.2.1 Behavioral Results

Table 3 presents QA accuracy across 7 languages. The results reveal three distinct behavioral patterns corresponding to intervention depth.

Shallow Layers. Consistent with our findings in the pilot study (Section 3.2), interventions at Layer 2 successfully reduce forget set accuracy to near-zero across all languages, but catastrophically degrade retain set performance for held-out languages (e.g., German: $58.0\% \rightarrow 1.0\%$, Italian: $62.0\% \rightarrow 0.0\%$), confirming that shallow layers encode fundamental multilingual representations.